KUSUM KUMARI

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EDUCATION

DAYANANDA SAGAR UNIVERSITY — **BANGALORE, INDIA**
Bachelor of Technology, Computer Science Engineering | CGPA: 9.16 — **2022-2026**

TECHNICAL SKILLS

- **Programming Languages & Web :** Java, JavaScript, Python, C, HTML,CSS
- **Frameworks & Libraries:** Spring Boot, React.js, Node.js, Express.js, Tailwind CSS
- **Databases, Tools & Cloud**: MySQL, Git, GitHub,AWS,Postman
- **Machine Learning & Data Science:** Scikit-learn, Pandas, NumPy, PyCaret, NLP (VADER)
- **Core Computer Science :** DSA, OOP, Operating System,Computer Networks, DBMS

PROJECTS

Unfair Trading Detection System-A Hybrid Approach | Python | PyCaret | RapidMiner | VADER | FDR | AutoML
A **Sentiment-Augmented Machine Learning Framework** for Detecting Anomalies in Financial Trading.

- Designed a hybrid anomaly detection system using statistical inference **(p-values, FDR**) and sentiment analysis (**VADER**) to improve trade classification accuracy.
- Developed and fine-tuned **Decision Tree** and **Random Forest** models in **RapidMiner**; automated model selection and optimization using **PyCaret AutoML**
- Minimized false positives and enhanced model precision by integrating real-time sentiment signals and hypothesis-driven feature engineering.

ExpenseMate | Spring Boot | MySQL | Spring Security | Thymeleaf | Chart.js
A Smart Expense & Budget Tracker with **gamified progress** and **automated spending alerts**

- Built a **secure role-based system** (User/Admin) using Spring Security with complete expense and budget management modules.
- Implemented **email alert automation** via Spring Scheduler & JavaMailSender for real-time budget notifications.
- Designed **interactive dashboards** and a **badge-based gamification system** to motivate consistent saving habits.

AgroLens | AI-Driven Crop Disease Forecast & Farmer Advisory| Python(ML) | Node.js | React.js | Firebase
An AI-powered platform that predicts crop disease outbreaks and provides actionable advisories to farmers using real-time environmental and satellite data.

- Built a full-stack web application integrating **React.js**, **Node.js**, and a **Flask-based AI microservice** for real-time predictive analytics.
- Trained and deployed machine learning models to forecast crop disease risks with region-specific insights, enabling early detection and improved decision-making for farmers.
- Implemented **Firebase authentication** and optimized cloud deployment to ensure secure and scalable user access.

ACHIEVEMENTS

NASA Space Apps Challenge 2024 – Global Nominee
Team: The Beginners | Challenge: Chronicles of Exoplanet Exploration

- Built a 3D interactive web app using React Three Fiber and Three.js to educate users about exoplanets through immersive visuals and a quiz-based learning experience, designed even for young learners.
- Selected as a Global Nominee from the Harohalli event for innovative solution development and efficient project delivery.
  Profile & Certificate: View Certificate & Team Profile

CERTIFICATIONS

- **Career Essentials in Software Development | Microsoft & LinkedIn**|2025
- **Data Visualization with Python | CognitiveClass.ai & IBM Developer Skills Network** | 2025
- **AWS Cloud Foundations** | **Amazon Web Services** | 2024

PUBLICATIONS &RESEARCH

- **Unfair Trading Detection System** — Research paper accepted for presentation at ***IEEE CSITSS-2025, Bengaluru***.